Sprint Planning: Prolecom Project

Final Optimized Version (Software Engineering II)

Prolecom Development Team

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1 Introduction

This document details the final technical planning for the Prolecom system. The structure is based on a 12-week agile cycle (4 sprints), integrating a rigorous Engineering Excellence approach: static code analysis, CI/CD pipelines, and a comprehensive testing plan to ensure the stability of the final product.

2 Sprint Schedule

2.1 Sprint 1: Foundations, CI/CD, and Security (Weeks 1-3)

Objective: Establish the core infrastructure and the permissions system (RBAC).

ID	User Story	Quality and Implementation Tasks
PB1	Infrastructure & CI/CD	Docker and GitHub Actions setup. Configuration of **SonarQube/ESLint** for static analysis.
PB2	Database	ER design and SQL/ORM migrations. Unit testing of models.
PB3	Auth & RBAC	JWT implementation for 6 roles. Route security testing.
PB4	Layout & Courses	Tailwind UI setup, main layout, and initial course CRUD.
QA	Baseline Analysis	Initial coverage report and role security validation.

2.2 Sprint 2: Learning Content and Challenge Repository (Weeks 4-6)

Objective: Enable the student's primary interaction with educational content.

ID	User Story	Quality and Implementation Tasks
PB6	Enrollment	Enrollment logic and filtering. Student-course integration testing.
PB13	Learning Materials	PDF and video uploads. Secure document viewer implementation.
PB8	Challenge Structure	Exercise bank (3 difficulty levels). Hidden test case validation.
STG	Staging Deployment	**Smoke testing** in a real environment. UI/UX bug fixes.

2.3 Sprint 3: Visual IDE, Collaboration, and Progress Tracking (Weeks 7-9)

Objective: Implement the coding environment (Monaco) and the academic forum.

ID	User Story	Quality and Implementation Tasks
PB9	Visual IDE (Monaco)	Monaco Editor integration. Syntax highlighting and secure environment.
PB12	Academic Forum (Q&A)	Course-based discussion threads. Load testing for concurrency.
PB16	Validation	Official answer labels (TAs/Instructors) and mentoring features.
PB11	Progress Tracking	Progress algorithms. Unit tests for completion calculations.

2.4 Sprint 4: Administration and Release (Weeks 10-12)

Objective: Complete administrative modules and ensure production readiness.

ID	User Story	Quality and Implementation Tasks
PB19	Assessments	Quizzes and flashcards. Automated grading tests.
PB17	Moderation	Reporting system and moderator dashboard (banning/hiding content).
PB22	Support & Admin	Account management and system health log monitoring.
PB23	Final Release	Complete **End-to-End (E2E) testing**. Final code audit.

3 Definition of Done (DoD)

No task will be considered complete until it meets the following criteria:

- **Static Analysis:** Code is free of critical issues in SonarQube/ESLint.
- **Testing:** Unit test coverage is validated and integration tests have passed.
- **Review:** *Code Review* approved by at least one team member.
- **CI/CD:** Successful and verified deployment in the Staging environment.

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Time Stamping
Security Data

Authorized Client Signature
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